12V DC Battery Power to 220V AC Converter

Project Report submitted
in partial fulfillment of requirements for the
6th Semester Mini Project

Bachelor of Technology
In
Electrical Engineering

By

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Declaration

We, the students of project group code: **PGEE363**, hereby declare that this project work titled **"12V DC Battery Power to 220V AC Converter"** is carried out by us in the Department of Electrical Engineering of Shri Guru Gobind Singhji Institute of Engineering & Technology, Vishnupuri, Nanded. The work is original and has not been submitted earlier whole or in part for the award of any degree/diploma at this or any other Institution / University.

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This is to certify that the project titled **"12V DC Battery Power to 220V AC Converter"**, submitted by **Krishna Bhise, Shrinivas Hamand, Satvik Burande, Darshan Singh, Mrunal Bhamare and Rupesh Pawar** in partial fulfillment of the requirements for the 6th-Semester Mini Project of **Bachelor of Technology in Electrical Engineering**, SGGSIE&T, Nanded. The work is comprehensive, complete and fit for final evaluation.

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ABSTRACT

The ever-increasing demand for uninterrupted electrical power, particularly in rural and off-grid environments, has necessitated the development of efficient and cost-effective power conversion systems. This project presents the design, development, and implementation of a 12V DC Battery Power to 220V AC Converter, commonly known as a DC-to-AC inverter, powered by a solar charging circuit.

The primary objective of this project is to construct a reliable, standalone inverter system capable of converting the 12V DC energy stored in a lead-acid battery into 220V AC power suitable for operating low-power household and electronic appliances. The system is designed without the use of microcontrollers, relying entirely on analog and power electronic components, thereby ensuring simplicity, low cost, and ease of maintenance.

The system is structured around three functional stages. The first is the Solar Charging Circuit, which uses an LM317 voltage regulator in conjunction with appropriate resistors, a Schottky diode for reverse current protection, and a 12V solar panel to safely charge the lead-acid battery. This stage ensures that the battery is maintained at the correct charging voltage and protected from overcharging.

The second stage is the PWM Inverter Circuit, which forms the core of the project. The SG3525 Pulse Width Modulation (PWM) controller IC generates a 50 Hz square-wave switching signal. This signal is used to drive two IRF520 N-channel Power MOSFETs in a push-pull configuration. The alternating switching of these transistors creates a low-voltage square-wave AC signal from the 12V DC battery supply.

The third and final stage is the Step-Up Transformer stage. A centre-tapped 12-0-12 V step-up transformer receives the MOSFET output and boosts it to 220V AC. An AC capacitor is used as a filter to smooth the output waveform and improve the quality of the delivered power.

The system was validated through breadboard prototyping, oscilloscope measurement of waveforms at critical points (MOSFET output, transformer primary and secondary), and practical load testing with a 5-watt LED bulb. The results confirmed correct operation of each stage, with a usable 220V AC output successfully powering small loads. A custom single-layer PCB was subsequently designed using KiCad and fabricated to provide a permanent and robust implementation.

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Nomenclature

- PWM – Pulse Width Modulation
- MOSFET – Metal Oxide Semiconductor Field Effect Transistor
- LDR – Light Dependent Resistor
- IC – Integrated Circuit
- PCB – Printed Circuit Board
- AC – Alternating Current
- DC – Direct Current

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CHAPTER 1:
INTRODUCTION

Introduction

The availability of clean and reliable electrical power is a fundamental requirement for modern life. In developing regions, remote areas, and during power outages, conventional electricity from the utility grid is often unavailable or unreliable. In such situations, standalone power systems that can convert stored DC energy into usable AC power become critically important. A DC-to-AC inverter is the key component of any such system, enabling batteries charged by renewable sources such as solar panels to power standard AC appliances.

This project presents the design and implementation of a 12V DC Battery Power to 220V AC Converter. The system is designed to take energy stored in a standard 12V lead-acid battery and convert it to a 220V, 50 Hz alternating current output capable of powering low-power household devices such as LED bulbs, mobile chargers, Wi-Fi routers, CCTV cameras, and emergency lights.

The core innovation of this design lies in its simplicity and accessibility. Rather than relying on complex digital microcontrollers, the inverter is built entirely from analog and power electronic components. The heart of the inverter circuit is the SG3525 PWM (Pulse Width Modulation) controller IC. This IC is specifically designed for switching power supply applications and generates a stable, adjustable-frequency square-wave output. In this project, it is configured to produce a 50 Hz switching frequency, which drives two IRF520 N-channel Power MOSFETs in a push-pull configuration.

The push-pull MOSFET stage drives the primary of a centre-tapped step-up transformer. This transformer boosts the 12V square-wave input to 220V AC at the secondary. An AC filter capacitor is placed at the output to improve waveform quality and provide a smoother voltage to the connected load.

To make the system self-sufficient, a solar charging circuit is integrated. A 12V solar panel charges the lead-acid battery through a regulated charging circuit built around the LM317 adjustable voltage regulator. This circuit controls the charging voltage, protecting the battery from overcharging, and includes a Schottky diode to prevent reverse current flow from the battery back to the solar panel at night.

The complete system was developed through a phased engineering workflow: initial circuit design and simulation, breadboard prototyping and testing, waveform verification using an oscilloscope, and final implementation on a custom-designed single-layer PCB using KiCad. This project not only demonstrates the practical application of power electronics but also provides valuable hands-on experience in analog circuit design, PCB layout, and hardware implementation.

CHAPTER 2:
LITERATURE REVIEW

Literature Review

The development of DC-to-AC power inverters has been a subject of active research and engineering for several decades, driven by the growing need for reliable off-grid and backup power systems. The evolution of inverter technology reflects advances in semiconductor devices, control ICs, and magnetic components.

Early inverter designs relied on electromechanical rotary converters and, later, on simple transistor-based switching circuits. These systems were often inefficient and prone to mechanical failure. The introduction of power MOSFETs in the 1980s marked a significant advancement, offering fast switching speeds, low on-state resistance, and ease of control with low gate-drive power requirements. Research by Mohan, Undeland, and Robbins in Power Electronics: Converters, Applications, and Design establish the foundational principles of PWM inverter design using power semiconductor switches, which forms the theoretical basis of this project.

The SG3525A PWM controller, introduced by Silicon General (now Texas Instruments), became a widely adopted IC for designing switching power supplies and inverters. Its application in push-pull and half-bridge topologies has been extensively documented in application notes and academic literature. The IC provides internal voltage reference, oscillator, error amplifier, and gate driver outputs, simplifying the design of complete inverter circuits around a single chip.

The push-pull topology employed in this project is one of the most commonly used configurations for low-power inverter applications. It uses a centre-tapped transformer with two switching transistors alternately conducting to create an AC waveform. Academic studies comparing inverter topologies — including push-pull, half-bridge, and full-bridge — consistently indicate that the push-pull configuration is most suitable for low-voltage, battery-powered applications due to its simplicity and low component count.

Regarding solar charging circuits, extensive literature exists on the use of linear regulators such as the LM317 for simple, cost-effective battery charging. While more advanced Maximum Power Point Tracking (MPPT) controllers offer greater efficiency, the LM317-based approach provides a straightforward and reliable constant-voltage charging method suitable for small-scale educational projects and low-power applications. The use of Schottky diodes to prevent reverse current discharge is a standard protection technique described in battery management literature.

Several undergraduate and published projects have demonstrated similar inverter systems. A notable observation across these works is the universal importance of correct transformer selection — specifically the turns ratio and core saturation characteristics — for achieving reliable voltage step-up. Additionally, output filtering using capacitors to reduce harmonic content in the output waveform is a recommended practice for improving power quality, as discussed in related IEEE publications on low-power inverter design.

Overall, the literature confirms that a reliable and functional DC-to-AC inverter for low-power applications can be effectively constructed using the SG3525 PWM controller, power MOSFETs, a centre-tapped step-up transformer, and appropriate passive components, without the complexity of microcontroller-based digital control. This project follows the same established approach, adapting proven design principles into a complete, fabricated hardware system.

CHAPTER 3:
Problem Statement & Objectives

Problem Statement

In numerous residential, agricultural, and institutional settings across developing regions, consistent access to 220V AC grid power is either unavailable or subject to frequent outages. During such outages, users are left without power for essential low-power devices including lighting, communication equipment, and small appliances. While large-scale UPS systems and generator sets exist, they are prohibitively expensive, require specialized maintenance, and are unsuitable for small-scale or educational applications.

Simultaneously, solar energy represents an abundant and freely available renewable resource. However, solar panels produce low-voltage DC power that cannot directly power standard 220V AC appliances. The energy must first be stored in a battery and then converted from DC to AC at the appropriate voltage and frequency.

The core problem addressed by this project is the absence of a simple, low-cost, hardware-only system that can:

- Safely capture solar energy and store it in a 12V lead-acid battery using a regulated charging circuit.
- Convert the stored 12V DC battery energy into usable 220V, 50 Hz AC power.
- Power basic low-wattage AC appliances (up to approximately 20W) reliably and without requiring microcontroller programming or complex digital control.

Objectives

The primary objectives of this project are designed to solve the identified problems through a complete, hardware-based engineering solution:

- To design and develop a solar-powered battery charging circuit using the LM317 voltage regulator that maintains safe and regulated charging of a 12V lead-acid battery from a solar panel.
- To design and implement a DC-to-AC inverter circuit using the SG3525 PWM controller IC that generates a stable 50 Hz switching signal.
- To drive two IRF520 Power MOSFETs in a push-pull configuration to amplify the PWM switching signal using stored battery energy.
- To use a centre-tapped 12-0-12 step-up transformer to boost the MOSFET output to 220V AC, with an AC filter capacitor for output smoothing.
- To validate the complete system through breadboard prototyping, oscilloscope measurement of waveforms at MOSFET and transformer outputs, and practical load testing with a 5W LED bulb.
- To implement the verified design on a custom-designed single-layer PCB using KiCad for a robust, permanent hardware solution.

CHAPTER 4:
System Methodology

System Methodology

The development of the 12V DC to 220V AC Converter followed a systematic and structured methodology, progressing from conceptual design through simulation, prototyping, testing, and final PCB fabrication. The system is divided into three major functional stages: the Solar Charging Stage, the PWM Inverter Stage, and the Step-Up Transformer Stage.

Solar Charging Stage

This stage is responsible for harvesting energy from the solar panel and safely storing it in the 12V lead-acid battery. The 12V solar panel generates a varying DC voltage depending on sunlight intensity. This raw solar output is fed through a Schottky diode (rated 3A, 50V) to prevent reverse current flow during low-light or nighttime conditions. The voltage is then regulated to a safe charging level using the LM317 adjustable voltage regulator. The output voltage of the LM317 is set using a combination of fixed resistors (220Ω and 680Ω) and a 2kΩ potentiometer, allowing fine adjustment of the charging voltage to match the battery's requirements. A 0.1 µF capacitor is placed at the input to filter noise. This stage ensures that the battery is charged efficiently and protected from overcharging.

PWM Inverter Stage

This is the core of the inverter system. The SG3525 PWM controller IC generates a stable 50 Hz square-wave output at its two complementary output pins (OUT A and OUT B). These outputs alternately drive the gates of two IRF520 N-channel Power MOSFETs. The oscillator frequency of the SG3525 is set using external resistors (10kΩ timing resistors and a 50kΩ variable resistor for frequency adjustment) and a 1µF timing capacitor. Gate resistors (47Ω) are placed in series with the MOSFET gates to limit switching transients and suppress oscillations. The two MOSFETs switch alternately, drawing current from opposite halves of the transformer primary in a push-pull configuration.

Step-Up Transformer Stage

A centre-tapped 12-0-12 V, 5A step-up transformer receives the alternating switched current from the MOSFET stage at its primary winding. As the two MOSFETs switch alternately, the transformer primary sees a square-wave AC signal. The transformer steps this voltage up to approximately 220V AC at its secondary. An AC filter capacitor is connected across the secondary output to suppress high-frequency harmonics and smooth the output waveform. The output is then available at the secondary terminals for connection to the load.

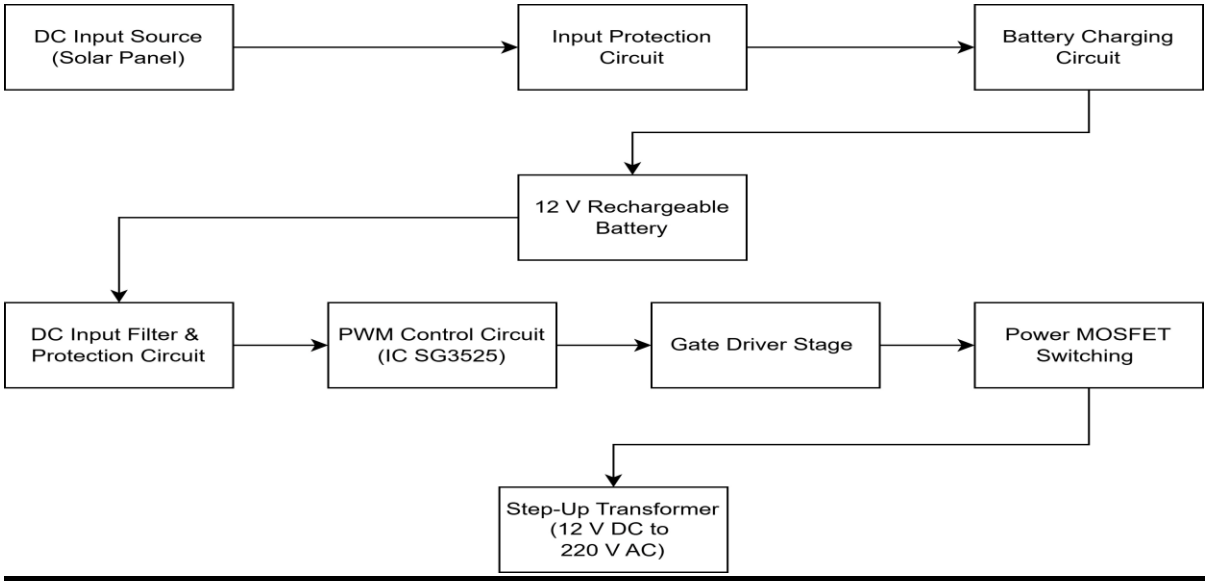
Development Workflow

The project followed a phased development approach:

- Research & Design: Literature study, component selection, and schematic design using Circuit Maker.
- Simulation: Functional verification of the PWM and MOSFET stages to confirm correct frequency and switching behaviour.
- Breadboard Prototyping: Assembly of both the charging circuit and inverter circuit on a breadboard for initial testing.
- Testing & Measurement: Oscilloscope measurement of waveforms at MOSFET output (drain), transformer primary, and secondary (220V AC output). Load testing with a 5W bulb.
- PCB Design: Single-layer PCB layout created in KiCad, with careful attention to track width, component placement, and heat dissipation for the power MOSFETs.
- PCB Fabrication & Assembly: PCB printed, drilled, and all components soldered. Final circuit validated against the original schematic.

CHAPTER 5:
Block Diagram & Working

Block Diagram



[Figure-1 Block Diagram]

The system block diagram illustrates the sequential flow of energy and signals from the solar panel through to the AC output. The solar panel feeds into the charging circuit, which regulates the voltage and charges the 12V lead-acid battery. The battery supplies DC power to the SG3525 PWM controller and to the drain terminals of both power MOSFETs. The SG3525 drives the gate terminals of the MOSFETs with complementary 50 Hz PWM signals. The MOSFET outputs drive opposite ends of the centre-tapped transformer primary. The secondary of the transformer produces 220V AC, filtered by an output capacitor before delivery to the load.

Working

The working principle of the 12V DC to 220V AC Converter is based on rapid, controlled switching of DC power through a transformer to produce an alternating output voltage. The system operates as follows:

1. Solar Charging Operation

The 12V solar panel converts incident sunlight into DC electrical energy. This energy is routed through a Schottky diode to prevent reverse current and passed through the LM317 voltage regulator circuit, which reduces and stabilizes the voltage to the safe charging level for the lead-acid battery. The battery gradually charges and stores the electrical energy for use by the inverter.

2. PWM Signal Generation

The SG3525 IC is powered directly from the battery. Its internal oscillator — configured by external R and C components — generates complementary square-wave pulses at 50 Hz on its OUT A and OUT B output pins. These pulses have a duty cycle close to 50%, meaning each output is HIGH for approximately one half-cycle and LOW for the other, with the two outputs never both HIGH simultaneously.

3. MOSFET Switching

The OUT-A signal from the SG3525 drives the gate of MOSFET 1 (Q1), and the OUT-B signal drives the gate of MOSFET 2 (Q2). When Q1 is ON, current flows from the battery through the upper half of the transformer primary and through Q1 to ground. When Q2 is ON, current flows from the battery through the lower half of the transformer primary and through Q2 to ground. This alternating current flow through the centre-tapped primary creates a push-pull AC signal in the transformer.

4. Transformer Step-Up

The centre-tapped 12-0-12 transformer receives the alternating 12V square-wave signal at its primary. Due to the turn's ratio of the transformer, the 220V AC is induced at the secondary winding. The output is available at 50 Hz, corresponding to the standard AC mains frequency.

5. Output Filtering and Load

An AC capacitor is connected across the secondary output to filter high-frequency harmonic content from the square wave, smoothing the output waveform. The filtered 220V AC output is then supplied to the connected load — in testing, a 5W LED bulb was used to confirm successful operation.

CHAPTER 6:
HARDWARE COMPONENTS

Hardware Components

The system comprises two sub-circuits: the Inverter Circuit and the Solar Charging Circuit. Each uses carefully selected components for reliable and safe operation. The complete component list is provided in the tables below.

Inverter Circuit Components

Sn no.	Components	Quantity
1	SG3525 IC	1
2	IRF520 MOSFET	2
3	Resistor 10KΩ	6
4	Resistor 4.7KΩ	1
5	Resistor 47Ω	1
6	Resistor 10Ω	2
7	Variable Resistor 50KΩ	1
8	Capacitor 1mF	1
9	Capacitor 1µF	2
10	Capacitor 100nF	1
11	LED 5mm Red	1
12	2 Pin Terminal Block	1
13	3 Pin Terminal Block	1
14	12V Lead Acid Battery	1
15	12-0-12 Transformer (5A)	1

Solar Charging Circuit Components

Sn no.	Components
1	Solar Panel – 12V
2	LM317 Voltage Regulator
3	DC Lead Acid Battery
4	Diode – 1N4007
5	Capacitor – 0.1µF
6	Schottky Diode – 3A, 50V
7	Resistors – 220Ω, 680Ω
8	Potentiometer – 2KΩ

Component Descriptions

1. SG3525 PWM Controller IC

The SG3525 is a dedicated PWM controller IC designed for switching power supply and inverter applications. It integrates an internal voltage reference (5.1V), a precision oscillator, an error amplifier, a soft-start circuit, and two complementary totem-pole output drivers. In this project, the SG3525 is configured to oscillate at 50 Hz by selecting appropriate timing resistors and a capacitor connected to its RT and CT pins. The complementary OUT A and OUT B outputs are used to drive the two power MOSFETs in the push-pull inverter stage. The adjustable output frequency allows fine-tuning via the variable 50kΩ resistor.

2. IRF520 N-Channel Power MOSFET

The IRF520 is an N-channel enhancement-mode power MOSFET with a drain-source voltage rating of 100V and a continuous drain current of 9.2A. Its low on-state resistance (RDS (on) = 0.27Ω) minimizes switching losses. In this project, two IRF520 devices are connected in a push-pull configuration, each switching the current through one half of the transformer primary. The gate is driven by the SG3525 outputs through 47Ω gate resistors to control switching speed and prevent oscillations.

3. Centre-Tapped Transformer (12-0-12 V, 5A)

The transformer is the key component for voltage step-up. The primary winding has a centre tap, with each half connected to one MOSFET drain. The secondary produces 220V AC output from the 12V DC input. The 5A current rating ensures that the transformer can handle the push-pull switching current without saturating. The turns ratio is carefully matched to achieve the desired 220V output.

4. LM317 Adjustable Voltage Regulator

The LM317 is a three-terminal adjustable positive voltage regulator that can supply up to 1.5A with an output voltage range of 1.25V to 37V. In the charging circuit, it is configured with fixed and variable resistors to set the battery charging voltage precisely. Its built-in current limiting and thermal shutdown features provide protection against overload and overheating during charging.

5. Schottky Diode (3A, 50V)

A Schottky diode is placed between the solar panel and the LM317 charging circuit to prevent reverse current from flowing from the battery back to the solar panel when panel voltage drops below battery voltage (e.g., at night or in shade). Its low forward voltage drops (typically 0.3V) minimizes power loss in the charging circuit.

6. 12V Lead-Acid Battery

A sealed lead-acid battery serves as the energy storage element. It stores the DC energy collected from the solar panel and supplies a steady 12V to the inverter circuit. Lead-acid batteries are chosen for this application due to their low cost, wide availability, and tolerance for deep discharge cycles in off-grid applications.

7. Passive Components (Resistors, Capacitors)

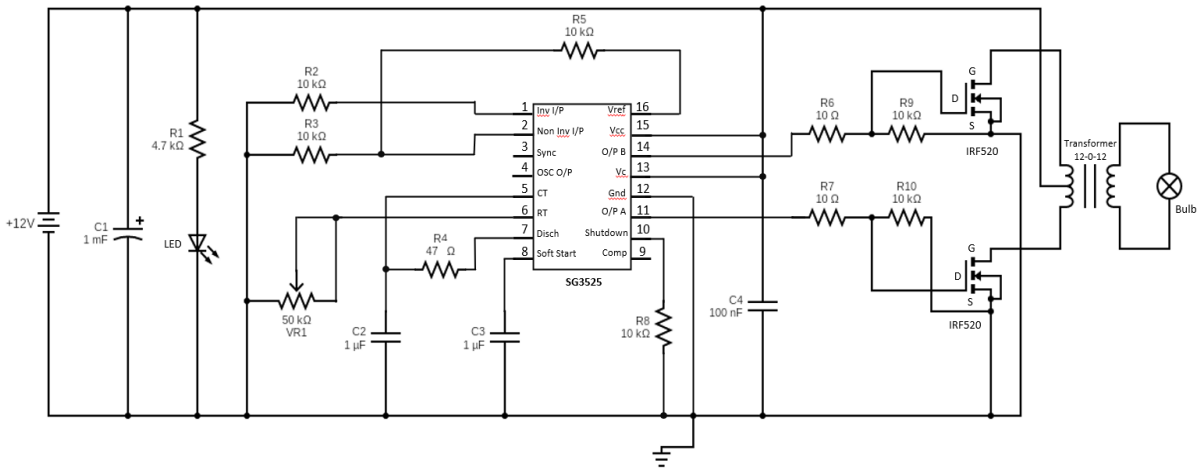
Various resistors are used for oscillator frequency setting, gate current limiting, and voltage division in the charging circuit. Capacitors serve multiple functions: the timing capacitor for the SG3525 oscillator, decoupling capacitors for stable IC operation, and the AC output filter capacitor at the transformer secondary to smooth the output waveform.

8. Printed Circuit Board (PCB)

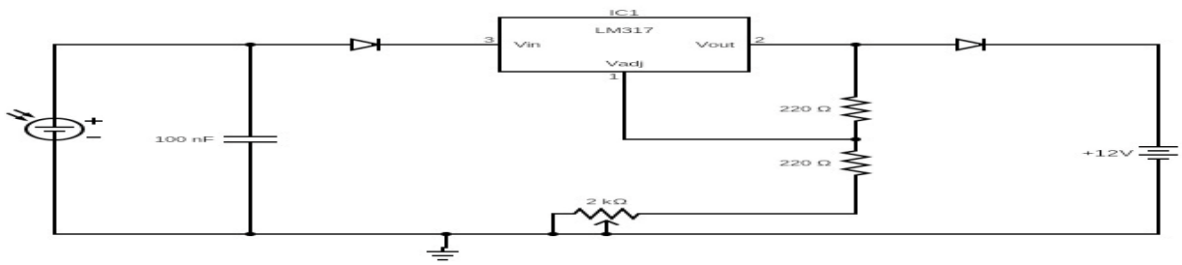
After successful breadboard testing, the circuit is implemented on a custom-designed single-layer PCB created using KiCad. The PCB provides permanent connections, better mechanical stability, reduced susceptibility to lose connections, and a compact layout suitable for a final product. Special attention is paid to trace widths for high-current paths (MOSFET drain and battery lines) and thermal management for the power devices.

CHAPTER 7:
Circuit Design & Implementation

Circuit Design



[Fig-2 Inverter Circuit Diagram]



[Fig-3 Solar Charging Circuit Diagram]

The circuit design of the 12V DC to 220V AC Converter is divided into two interconnected sub-circuits: the Solar Charging Circuit and the PWM Inverter Circuit. The complete schematic was drafted using Circuit Maker software to verify connections, component values, and overall circuit topology before physical assembly.

In the Inverter Circuit, the SG3525 IC is the control hub. Its oscillator is configured with R(T) = 10kΩ and C(T) = 1μF to target a 50 Hz output frequency. The DEAD TIME pin is connected to a 4.7kΩ resistor to set a small dead-time between complementary outputs, preventing simultaneous conduction of both MOSFETs (shoot-through). The SOFT START pin is connected to a 1μF capacitor for controlled start-up. The two complementary outputs (OUT A and OUT B) each drive one MOSFET gate through a 47Ω series resistor. The MOSFET drain terminals connect to opposite ends of the centre-tapped transformer primary, with the centre tap connected to the positive terminal of the battery.

In the Solar Charging Circuit, the LM317 output voltage is set by the resistor network (R1 = 220Ω, R2 = 680Ω, and a 2kΩ potentiometer in series) according to the formula: Vout = 1.25 × (1 + (R2/R1)). The potentiometer allows fine adjustment of the charging voltage. The Schottky diode is placed in series with the solar panel output to block reverse current. A 0.1μF capacitor is placed at the LM317 input for stability.

Implementation

The implementation of the project followed a well-defined and structured workflow to ensure systematic progress from design to final hardware.

The project began with a thorough study of the SG3525 PWM controller datasheet and application notes for push-pull inverter design, as well as the LM317 datasheet for the solar charging circuit. Component values were calculated and verified through analysis of the SG3525 oscillator frequency equation and the LM317 output voltage formula. The complete schematic was then drawn and checked in Circuit Maker.

All required components were procured and the circuit was first assembled on a breadboard. The charging circuit was tested independently using a bench power supply to simulate the solar panel input and verify correct output voltage at the LM317. The inverter circuit was then assembled on the breadboard. An oscilloscope was used to probe the SG3525 output pins to verify the 50 Hz complementary switching waveforms, and then used at the MOSFET drain nodes to confirm the push-pull switching behaviour.

With the breadboard circuit functioning correctly, the PCB layout was designed in KiCad. Critical design decisions during PCB layout included: using wider tracks (minimum 2mm) for the high-current battery and MOSFET drain connections; placing decoupling capacitors close to the SG3525 IC supply pin; and providing adequate spacing around the IRF520 MOSFETs to allow heat dissipation. The single-layer PCB was then printed, drilled, and components were carefully soldered.

After PCB assembly, the circuit was powered up progressively: first the SG3525 stage alone, then with the MOSFETs connected to the transformer, and finally with the transformer secondary connected to the filter capacitor and a 5W test load. Waveforms were re-verified at each stage to ensure correct operation. Work was distributed among team members to efficiently cover all phases of implementation.

Work Distribution

Task	Team Members
Circuit Design & Simulation	Mrunal Bhamare & Krishna Bhise
Breadboard Implementation	Satvik Burande & Darshan Singh
PCB Designing	Krishna Bhise, Satvik Burande, Shrinivas Hamand & Rupesh Pawar
Hardware Implementation & Testing	Shrinivas Hamand, Mrunal Bhamare, Darshan Singh & Rupesh Pawar

CHAPTER 8:
Results & Discussion

Results

The 12V DC to 220V AC Converter was developed, assembled, and tested through multiple stages — from breadboard prototyping to PCB fabrication — to systematically evaluate the performance and reliability of each sub-circuit and the integrated system.

Solar Charging Circuit Results

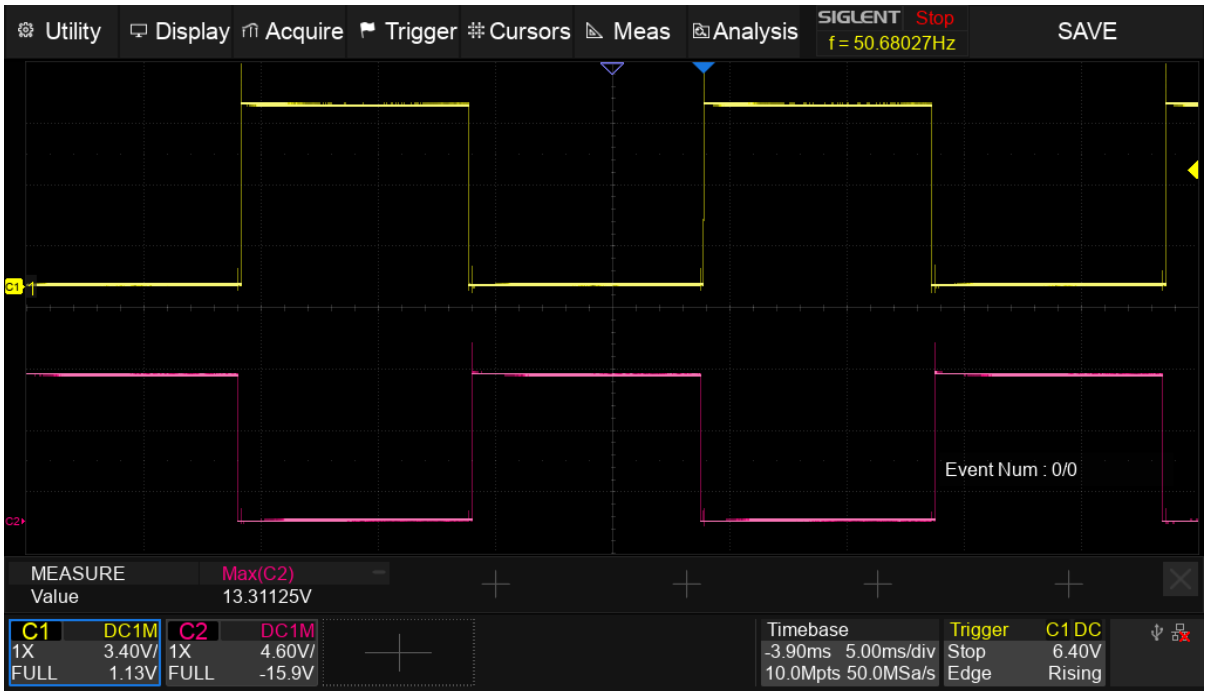
The LM317-based charging circuit performed as designed. When supplied with a simulated solar panel voltage of 14-16V, the circuit produced a regulated output voltage at the battery terminals, adjustable between 13.5V and 14.4V using the 2kΩ potentiometer. The Schottky diode effectively blocked reverse current in all tested conditions. The charging circuit successfully delivered current to the battery without overheating the LM317 for the tested input conditions.

SG3525 PWM Stage Results

The SG3525 PWM controller generated stable complementary 50 Hz square-wave outputs on its OUT A and OUT B pins, as verified on the oscilloscope. The waveforms showed the correct dead-time between complementary signals, confirming that both outputs were never simultaneously HIGH, preventing cross-conduction. The variable resistor allowed fine adjustment of the switching frequency around the target 50 Hz.

MOSFET Switching Stage Results

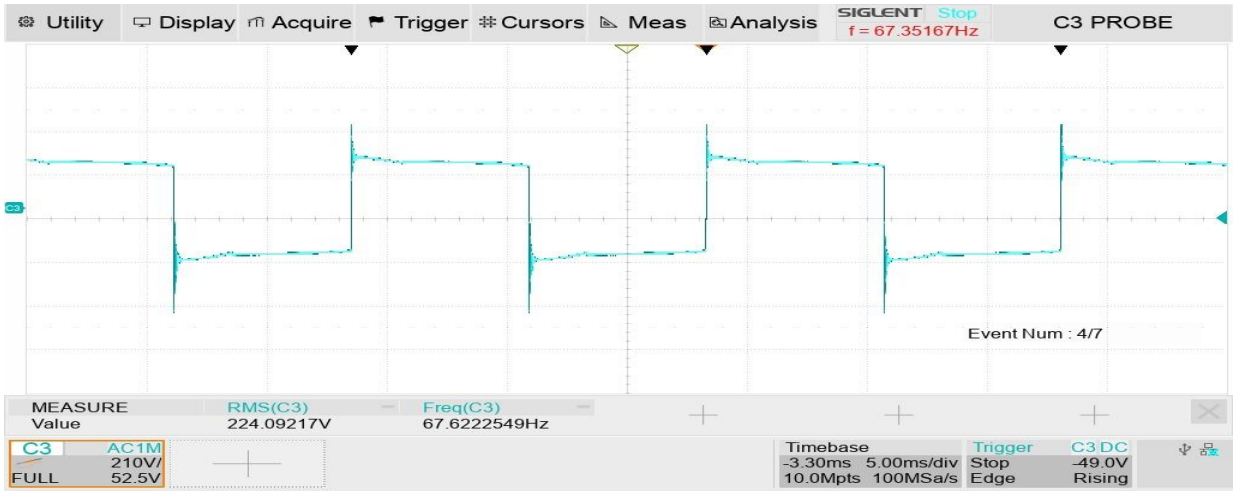
Oscilloscope measurements at the drain terminals of both IRF520 MOSFETs confirmed correct alternating switching behaviour. The MOSFET output waveform showed clean, rectangular transitions at 50 Hz, with both devices operating in complementary fashion. No simultaneous conduction was observed, validating the dead-time configuration of the SG3525.



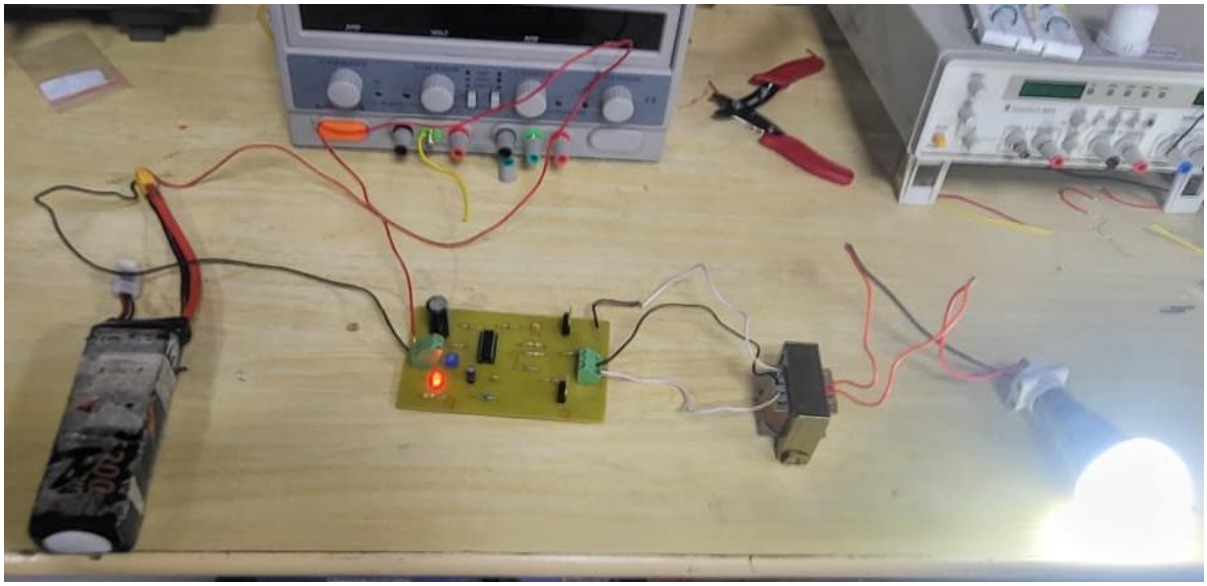
[Fig-4 MOSFET Output Waveform]

Transformer and AC Output Results

The centre-tapped transformer successfully stepped up the 12V push-pull signal to approximately 220V AC at its secondary terminals. Oscilloscope measurements at the transformer secondary showed a square-wave AC waveform at 50 Hz with a peak value consistent with 220V AC operation. The AC filter capacitor reduced high-frequency ripple at the output, improving waveform quality. The inverter successfully powered a 5W LED bulb, confirming practical output capability.



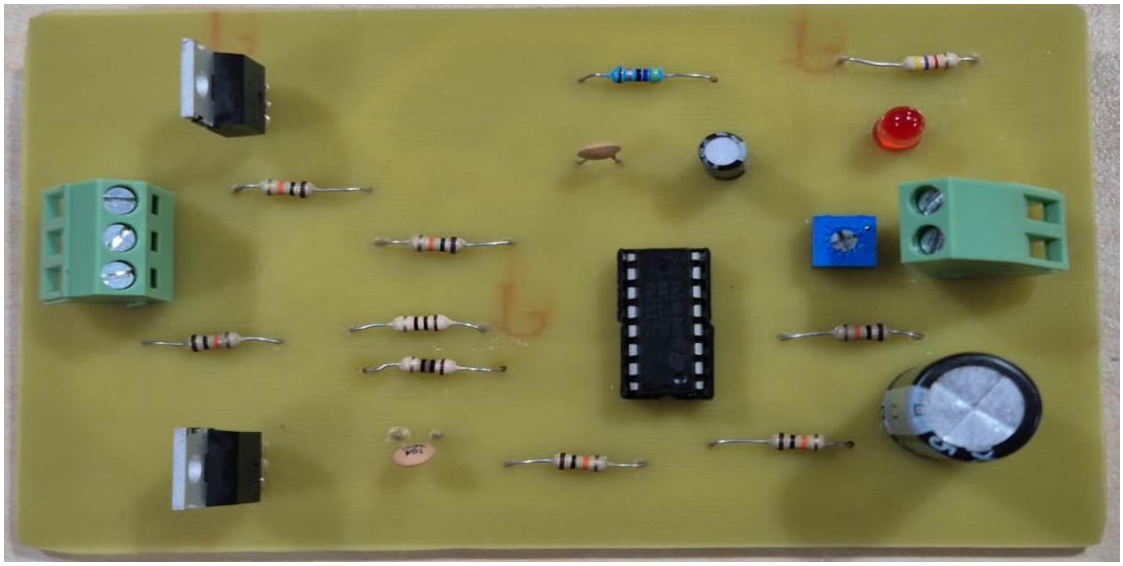
[Fig-5Transformer Output Waveform]



[Fig-6 Inverter Output with Load]

PCB Implementation Results

The custom-designed single-layer PCB was successfully fabricated and assembled. Basic circuit functionality — including SG3525 oscillation, MOSFET switching, and partial transformer output — was verified on the assembled PCB. Minor issues observed during initial PCB testing, such as incomplete output at certain load conditions, were attributed to soldering quality at high-current traces. These were identified and corrected through re-soldering and visual inspection.



[Fig-7 PCB]

Discussion

The results confirm that a fully functional DC-to-AC inverter can be built using simple, low-cost analog components without microcontroller involvement. The SG3525 IC proved to be a reliable and easy-to-configure PWM source, and the push-pull MOSFET stage effectively amplified the switching signal to drive the transformer.

The primary challenge encountered was in the PCB stage, where high-current paths required careful attention to trace width and soldering quality. This highlighted an important practical lesson: PCB design for power electronics must account for thermal and current-carrying constraints, not just electrical connectivity.

The output waveform, being a square wave rather than a pure sine wave, is suitable for resistive loads (LED bulbs, heating elements) and basic electronic devices with switching power supplies. It is not ideal for motor-driven appliances or sensitive equipment requiring pure sine wave input. This is a known limitation of simple push-pull inverter designs and represents an area for future improvement.

CHAPTER 9:
Conclusion & Future Scope

Conclusion

The 12V DC Battery Power to 220V AC Converter project successfully demonstrated the design and hardware implementation of a complete solar-powered inverter system using established analog and power electronic components. Through the integration of a solar charging circuit, an SG3525 PWM inverter stage, and a centre-tapped step-up transformer, the system achieved its primary objective of converting stored 12V DC battery energy into usable 220V, 50 Hz AC power.

The breadboard prototype confirmed correct operation of all three stages: regulated battery charging from the solar panel, stable 50 Hz PWM switching by the SG3525, and successful voltage step-up by the transformer to 220V AC. Practical load testing with a 5W LED bulb validated the real-world utility of the system. The PCB implementation further strengthened the team's understanding of power electronics hardware design, including the importance of trace width, thermal management, and soldering quality in high-current circuits.

Overall, the project achieved all its stated objectives and provided comprehensive hands-on experience in analog circuit design, power electronics, PCB layout using KiCad, waveform measurement using an oscilloscope, and systematic hardware debugging. It demonstrates how fundamental electronic components can be combined to create a practical, useful power system with real-world applications in off-grid and backup power scenarios.

Future Scope

While the current system provides a functional baseline inverter, several enhancements can significantly improve its performance, efficiency, safety, and range of applications:

1. Pure Sine Wave Output

The current square wave output can be replaced with a pure sine wave output by implementing a low-pass LC filter at the transformer secondary, or by redesigning the inverter around a

microcontroller or dedicated sine-wave PWM controller. This would allow the inverter to power sensitive electronics, motors, and appliances that require clean sinusoidal AC input.

2. MPPT Solar Charging

Replacing the simple LM317-based charging circuit with an MPPT (Maximum Power Point Tracking) charge controller would significantly improve energy harvesting efficiency, especially under variable sunlight conditions, resulting in faster battery charging and better utilization of available solar energy.

3. Increased Power Output

Scaling the inverter to higher power (100W–500W) by using higher-rated MOSFETs (e.g., IRF3205 or IRFZ44N), a more powerful transformer, and a larger battery bank would extend the range of applications to power fans, TVs, and larger appliances.

4. Battery Protection and Management

Adding a Battery Management System (BMS) with features such as over-discharge protection, over-charge protection, and state-of-charge (SOC) monitoring would extend battery life and improve system safety.

5. Microcontroller Integration

Incorporating a microcontroller such as an Arduino or ESP32 would enable intelligent features including automatic frequency adjustment, real-time monitoring of input voltage, output voltage, current, and power; automatic load shedding; and remote monitoring via a mobile app or web interface.

6. Automatic Voltage Regulation (AVR)

Implementing a feedback control loop around the SG3525 error amplifier input to monitor the output voltage and automatically adjust the PWM duty cycle would maintain a more stable output voltage across varying load conditions.

7. Improved PCB and Enclosure

A revised PCB with wider power traces, a proper ground plane, and heatsink provisions for the MOSFETs would improve thermal performance and reliability. Enclosing the system in a proper housing with ventilation would make it suitable for practical field deployment.

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